

VR SuperGun: Interfacing 1980s Arcade Hardware with Online Virtual Reality

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ABSTRACT

VR SuperGun is a custom hardware and software prototype allowing play of original arcade platforms through a network connection, reconstituting the material form of the arcade cabinet in digital space. It extends the format of the SuperGun, a device that contains the wiring of an arcade cabinet in consolised form.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Interaction design theory, concepts and paradigms*; • **Hardware** → **Emerging interfaces**.

KEYWORDS

virtual reality, webxr, physical computing, arcade, videogames, emulation

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1 INTRODUCTION

VR SuperGun is a unique working prototype that connects legacy arcade platforms to web based virtual reality environment through a custom hardware and software interface (figure 1).

It is an original design solution that has not been replicated since. There are several VR arcade environments that act as 3D shells for software based arcade emulators, including *EmuVR* [EmuVR 2018], *New Retro Arcade: Neon* [Cybercherries 2016], and *VR (Game Centre) Arcade Environment* [Noe 2021].

What makes *VR SuperGun* unique is that instead of using arcade emulation it directly connects to and interactively streams from physical arcade hardware into virtual reality space.

As an experimental prototype, *VR SuperGun* bridges the gap in-between the authenticity and tangibility of experiencing arcade gameplay through original hardware and in situated space around

the arcade cabinet, netplay, and the gameplay spectatorship afforded through livestreaming.

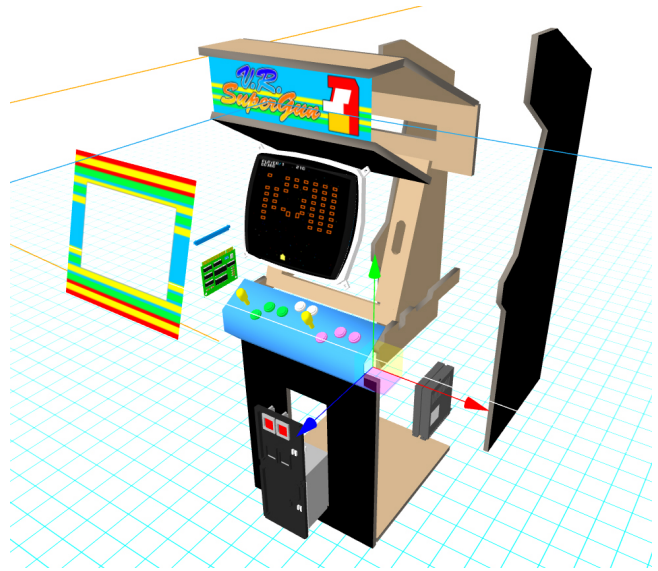


Figure 1: An exploded view of the virtual arcade cabinet playable in a web browser, with its virtual display streaming direct from a JAMMA standard arcade board.

2 DESIGN AND IMPLEMENTATION

The *VR SuperGun* prototype uses custom hardware and software. The hardware aspect augments a standard JAMMA SuperGun [Nolan 2019] device with a custom Arduino and optoisolator circuit allowing remote control from the connected PC server. The software aspect of this project is built largely upon JavaScript and open web standards, including A-Frame, Socket.IO, PhaserJS, Johnny-Five, and Node.js.

2.1 Arcade Video to VR Display

The connected arcade game board is connected to a standard SuperGun device supplying power, audiovisual outputs, and game control inputs that delivers a 240 line RGB video at 15khz. This low resolution analog video signal is streamed to the host PC by first passing through an upscaler that converts the analog signal to HDMI. The resulting HDMI signal is then sent to the host PC via a USB 3.0 game capture unit (figure 3). The video signal is then converted to a streaming HTML canvas texture via FFMpeg and

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custom Javascript, which is mapped to a 3D representation of an CRT arcade monitor inside the virtual environment [Nolan 2018].

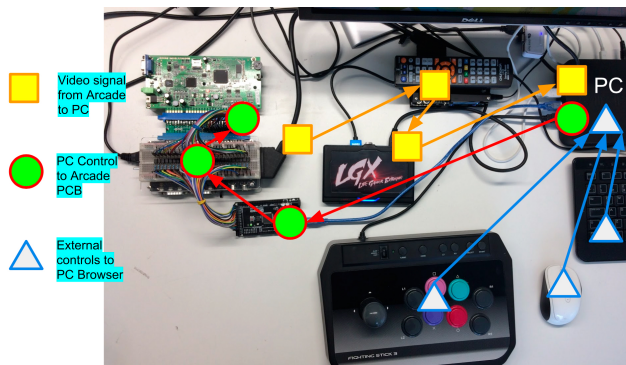


Figure 2: The VR SuperGun hardware setup (minus display).

2.2 VR to Arcade Hardware Controls

The game board is controlled from the A-Frame WebXR environment using the keyboard or joystick controls via Phaser.js. The current Oculus Quest implementation uses the headset’s bluetooth keyboard functionality to communicate with the WebXR browser.

The control inputs are sent to an Arduino Mega board connected to a custom optoisolator array mapped the 26 inputs on a JAMMA pass-through connector that bridges the game board and SuperGun, supporting 2 joysticks and 6 action buttons per player, along coin input, player start, and test input functionality (figure 2).

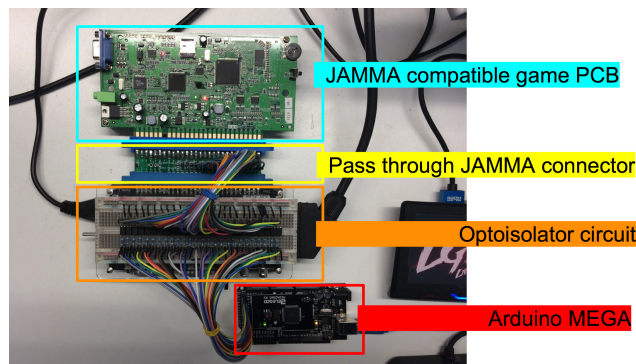


Figure 3: Arduino to JAMMA circuit detail.

2.3 VR Environment

The *VR SuperGun* prototype runs in A-Frame WebXR on a localhost HTTP server and is mirrored to secure remote https through Ngrok. A-Frame provides a scalable experience that's accessible in flat 3D on a standard desktop monitor, but also works with any WebXR compatible VR or MR headset. *VR SuperGun*'s low polygon visual style was chosen in honour of the era of 1980s and 1990s arcade gaming it is inspired by, while also optimising network speed and browser performance.

VR *SuperGun* presents the user with a 1:1 scale virtual reconstruction of a Dynamo HS-1 arcade cabinet (figure 1) [noa 2016], while also providing a view of the inner workings of the arcade machine as a historical and technical document of coin-op hardware design (figure 4).

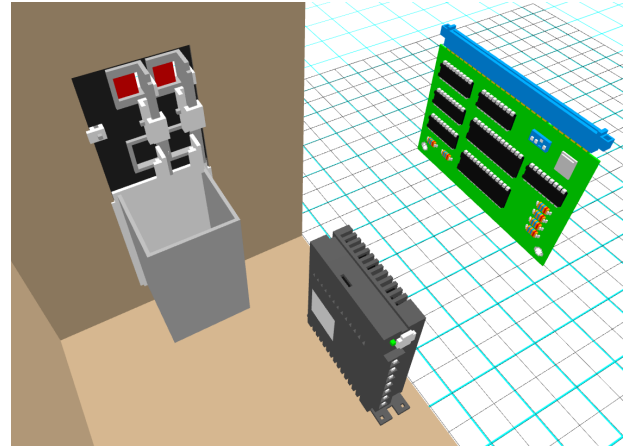


Figure 4: Inner detail of the virtual arcade cabinet, featuring the game PCB, power supply, and coin mechanism.

3 DISCUSSION

VR SuperGun brings the 1985 JAMMA hardware standard into networked virtual space. As a technical solution it delivers a trade off between the immediacy and accessibility of emulation, and the tangible and subjective authenticity experienced through using original arcade hardware.

The artefact also points a way for hybridising physical computational hardware and software simulation, by extending legacy videogame platforms into cyberspace. By embodying and reconstituting the physical presence of the arcade machine in digital 3D space *VR SuperGun* removes issues of perishability and impermanence, while opening up paths of discourse into how the authentic can be imbued into a digital reproduction and augmentation of actual reality.

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